Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In a relay race, there are six teams A, B, C, D, E and F. What is the probability that A, B, C finish first, second, third respectively?

(A) $\frac{1}{2}$

(B) $\frac{1}{12}$

(C) $\frac{1}{60}$

(D) $\frac{1}{120}$

Q.2

Two events A and B are such that $\mathrm{P}(\text{not } \mathrm{B}) = 0.8$, $\mathrm{P}(\mathrm{A} \cup \mathrm{B}) = 0.5$ and $\mathrm{P}(\mathrm{A} \mid \mathrm{B}) = 0.4$. Then $\mathrm{P}(\mathrm{A})$ is equal
to

(A) 0.28

(B) 0.32

(C) 0.38

(D) None of the above

Q.3

Assume that each born child is equally likely to be a boy or a girl. If a family has two children, then the conditional probabilities that both are girls given that (i) the youngest is a girl, (ii) at least one is a girl are

(A) $\frac{1}{2}$ and $\frac{1}{4}$

(B) $\frac{1}{3}$ and $\frac{1}{2}$

(C) $\frac{1}{3}$ and $\frac{1}{4}$

(D) $\frac{1}{2}$ and $\frac{1}{3}$

Q.4

One ticket is selected at random from 100 tickets numbered $00, 01, 02, \ldots, 98, 99$. If x_1 and x_2 denotes the sum and product of the digits on the tickets,
then $P\left(x_1 = 9 / x_2 = 0\right)$ is equal to

(A) $2/19$

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(B) $19/100$

(C) $1/50$

(D) None of these

Q.5

A die is rolled three times. Let E_1 denote the event of getting a number larger than the previous number each time and E_2 denote the event that the numbers (in order) form an increasing AP then

(A) $P\left(E_2\right) \geq P\left(E_1\right)$

(B) $P\left(E_2 \cap E_1\right)=\frac{3}{10}$

(C) $P\left(E_2 / E_1\right)=\frac{1}{36}$

(D) $P\left(E_1\right)=\frac{10}{3} P\left(E_2\right)$

Q.6

In a class 30% students like tea, 20% like coffee and 10% like both tea and coffee. A student is selected at random then what is the probability that he does not like tea if it is known that he likes coffee?

(A) $1/2$

(B) $3/4$

(C) $1/3$

(D) None of these

Q.7

Let E^{c} denote the complement of an event E. Let $\mathrm{E}, \mathrm{F}, \mathrm{G}$ be pairwise independent events with $\mathrm{P}(\mathrm{G})>0$ and $\mathrm{P}(\mathrm{E} \cap \mathrm{F} \cap \mathrm{G})=0$. Then $\mathrm{P}\left(\mathrm{E}^{\mathrm{c}} \cap \mathrm{F}^{\mathrm{c}} \mid \mathrm{G}\right)$
equals

(A) $\mathrm{P}\left(\mathrm{E}^{\mathrm{c}}\right)+\mathrm{P}\left(\mathrm{F}^{\mathrm{c}}\right)$

(B) $\mathrm{P}\left(\mathrm{E}^{\mathrm{c}}\right)-\mathrm{P}\left(\mathrm{F}^{\mathrm{c}}\right)$

(C) $\mathrm{P}\left(\mathrm{E}^{\mathrm{c}}\right)-\mathrm{P}(\mathrm{F})$

(D) $\mathrm{P}(\mathrm{E})-\mathrm{P}\left(\mathrm{F}^{\mathrm{c}}\right)$

Q.8

The probability of the simultaneous occurrence of two events A and B is p. If the probability that exactly one of the events occurs is q, then which of the following is not correct?

(A) $\mathrm{P}\left(\mathrm{A}^{\prime}\right)+\mathrm{P}\left(\mathrm{B}^{\prime}\right)=2+2 \mathrm{q}-\mathrm{p}$

(B) $\mathrm{P}\left(\mathrm{A}^{\prime}\right)+\mathrm{P}\left(\mathrm{B}^{\prime}\right)=2-2 \mathrm{p}-\mathrm{q}$

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(C) $\mathrm{P}(\mathrm{A} \cap \mathrm{B} \mid \mathrm{A} \cup \mathrm{B})=\frac{\mathrm{p}}{\mathrm{p}+\mathrm{q}}$

(D) $\mathrm{P}\left(\mathrm{A}' \cap \mathrm{B}'\right)=1-\mathrm{p}-\mathrm{q}.$

Q.9

A father has 3 children with at least one boy. The probability that he has 2 boys and 1 girl is

(A) $1/4$

(B) $1/3$

(C) $2/3$

(D) None of these

Q.10

it is given that the events A and B are such that $P(A)=1/4, P(A/B)=1/2$ and $P(B/A)=2/3$. Then $\mathrm{P}(\mathrm{B})$ is

(A) $1/3$

(B) $2/3$

(C) $1/2$

(D) $1/6$

Q.11

There are n letters and n addressed envelopes, the probability that all the letters are not kept in the right envelope, is

(A) $\frac{1}{n!}$

(B) $1-\frac{1}{n!}$

(C) $1-\frac{1}{n}$

(D) None of these

Q.12

A man has a bunch of 10 keys out of which only one can open the door. He choses a key at random for opening the door. If at each trial the wrong key is discarded, then the probability that the door is opened on fifth trial is

(A) $\frac{1}{2}$

(B) $\frac{10}{10^5}$

(C) $\frac{1}{10}$

(D) $\frac{5!}{10!}$

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Q.13

3 friends A, B and C play the game "Pahle Hum Pahle Tum" in which they throw a die one after the other and the one who will get a composite number 1st will be announced as winner, If A started the game followed by B and then C then what is the ratio of their winning probabilities?

(A) $9:6:4$

(B) $8:6:5$

(C) $10:5:4$

(D) None of these

Q.14

A is one of 6 horses entered for a race, and is to be ridden by one of two jockeys B and C. It is 2 to 1 that B rides A, in which case all the horses are equally likely to win. If C rides A, his chance of winning is trebled. What are the odds against winning of A ?

(A) $5:13$

(B) $5:18$

(C) $13:5$

(D) None of these

Q.15

A fair coin is tossed a fixed number of times. If the probability of getting seven heads is equal to that of getting nine heads, the probability of getting two heads is

(A) $\frac{15}{2^8}$

(B) $\frac{2}{15}$

(C) $\frac{15}{2^{13}}$

(D) None of these

Q.16

A die marked $1, 2, 3$ in red and $4, 5, 6$ in green is tossed. Let A be the event, "the number is even" and B be the event, "the number is red" then;

(A) $\mathrm{P}(\mathrm{A})\mathrm{P}(\mathrm{B}) = \frac{1}{6}$

(B) A and B are independent

(C) A and B are dependent

(D) None of these

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Q.17

The probability that A speaks truth is $\frac{4}{5}$, while this probability for B is $\frac{3}{4}$. The probability that they contradict each other when asked to speak on a fact is

(A) $\frac{3}{20}$

(B) $\frac{4}{5}$

(C) $\frac{7}{20}$

(D) $\frac{1}{5}$

Q.18

The probability of India winning a test match against Westindies is $\frac{1}{2}$ assuming independence from match to match the probability that in a 5 match series India's second win occurs at the third test, is

(A) $2/3$

(B) $1/2$

(C) $1/4$

(D) $1/8$

Q.19

If E_1 and E_2 are two events such that $\mathrm{P}\left(\mathrm{E}_1\right)=1/4\,\mathrm{P}\left(\mathrm{E}_2/\mathrm{E}_1\right)=1/2$ and $\mathrm{P}\left(\mathrm{E}_1/\mathrm{E}_2\right)=1/4$, then choose
the incorrect statement.

(A) E_1 and E_2 are independent

(B) E_1 and E_2 are exhaustive

(C) E_2 is twice as likely to occur as E_1

(D) Probabilities of the events $E_1 \cap E_2$, E_1 and E_2 are in G.P.

Q.20

Probability that a man who is 40 year old, living till 75 years is $5/16$, and another man who is 35 years old living till 70 years is $3/7$ then what is the probability that at least one of them will be alive till 35 years hence?

(A) $11/28$

(B) $19/28$

(C) $17/28$

(D) None of these

Answers & Solutions

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Q.1 Answer:

$\frac{1}{120}$

Solution:

Probability of $\mathrm{A} = \frac{1}{6}$
Probability of $\mathrm{B} = \frac{1}{5}$
Probability of $\mathrm{C} = \frac{1}{4}$

$$\text{Hence, required probability } = \frac{1}{6} \times \frac{1}{5} \times \frac{1}{4}$$
$$= \frac{1}{120}$$

Q.2 Answer:

0.38

Solution:

$\because \mathrm{P}(\overline{\mathrm{B}}) = 0.8 \Rightarrow \mathrm{P}(\mathrm{B}) = 0.2$
$\mathrm{P}(\mathrm{A} \cup \mathrm{B}) = 0.5 \& \mathrm{P}(\mathrm{A} \mid \mathrm{B}) = 0.4$
$\because \mathrm{P}(\mathrm{A} \cap \mathrm{B}) = \mathrm{P}(\mathrm{B}) \times \mathrm{P}(\mathrm{A} \mid \mathrm{B}) = 0.2 \times 0.4 = 0.08$
$\& \mathrm{P}(\mathrm{A}) = \mathrm{P}(\mathrm{A} \cup \mathrm{B}) - \mathrm{P}(\mathrm{B}) + \mathrm{P}(\mathrm{A} \cap \mathrm{B})$
$\mathrm{P}(\mathrm{A}) = 0.5 - 0.2 + 0.08 = 0.38$

Q.3 Answer:

$\frac{1}{2}$ and $\frac{1}{3}$

Solution:

Let b **and** g **represent the boy and the girl child, respectively. If a family has two children, the sample space will be** $S = \{(b, b), (b, g), (g, b), (g, g)\}$
Let A **be the event that both children are girls. Therefore,** $A = \{(g, g)\}$
(i) Let B **be the event that the youngest child is a girl. Therefore,** $B = [(b, g), (g, g)] \Rightarrow A \cap B = \{(g, g)\}$
$\therefore P(B) = \frac{2}{4} = \frac{1}{2}$ **and** $P(A \cap B) = \frac{1}{4}$
The conditional probability that both are girls, given that the youngest child is a girl, is given by $P(A/B)$
$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{1/2} = \frac{1}{2}$
(ii) Let C **be the event that at least one child is a girl. Therefore,** $C = \{(b, g), (g, b), (g, g)\}$
$\Rightarrow A \cap C = \{g, g\} \quad \Rightarrow P(C) = \frac{3}{4}$ **and**
$P(A \cap C) = \frac{1}{4}$
The conditional probability that both are girls, given that at least one child is a girl, is given by $P(A \mid C)$**. Therefore,** $P(A \mid C) = \frac{P(A \cap C)}{P(C)} = \frac{1/4}{3/4} = \frac{1}{3}$.

Q.4 Answer:

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$2/19$

Solution:

Let the number selected be xy. Then $x+y=9, 0 \leq x, y \leq 9$ and

$xy=0 \Rightarrow x=0, y=9$

or $\quad y=0, x=9$

$P\left(x_1=9 / x_2=0\right)=\frac{P\left(x_1=9 \cap x_2=0\right)}{P\left(x_2=0\right)}$

Now, $P\left(x_2=0\right)=\frac{19}{100}$ **and**

$P\left(x_1=9 \cap x_2=0\right)=\frac{2}{100}$

$\Rightarrow P\left(x_1=9 / x_2=0\right)=\frac{2 / 100}{19 / 100}=\frac{2}{19}.$

Q.5 Answer:

$P\left(E_1\right)=\frac{10}{3} P\left(E_2\right)$

Solution:

$P\left(E_1\right)=\frac{{}^6C_3}{216}=\frac{5}{54}; P\left(E_2\right)=\frac{4+2}{216}=\frac{1}{36}$

$E_1 \cap E_2=E_2 \quad \therefore P\left(E_1 \cap E_2\right)=P\left(E_2\right)=1/36$

$P\left(E_2 / E_1\right)=\frac{P\left(E_1 \cap E_2\right)}{P\left(E_1\right)}=\frac{1/36}{5/54}=\frac{54}{180}=\frac{3}{10}$

Q.6 Answer:

$1/2$

Solution:

Let $\mathrm{P}(\mathrm{A})=$ **probability that a randomly selected student likes tea** $=0.3$ **Let** $\mathrm{P}\left(\mathrm{A}_2\right)=$ **probability that a randomly selected student does not like tea** $=1-0.3=0.7$ **Let** $\mathrm{P}(\mathrm{B})=$ **probability that a randomly selected student likes coffee** $=0.2$

$\therefore \mathrm{P}\left(\mathrm{A}_2 \cap \mathrm{B}\right)=\mathrm{P}(\mathrm{B})-\mathrm{P}(\mathrm{A} \cap \mathrm{B})=0.2-0.1=0.1$

Now we have to find $\mathrm{P}\left(\mathrm{A}_2 / \mathrm{B}\right)=\mathrm{P}\left(\mathrm{A}_2 \cap \mathrm{B}\right) / \mathrm{P}(\mathrm{B})=0.1 / 0.2=1/2$

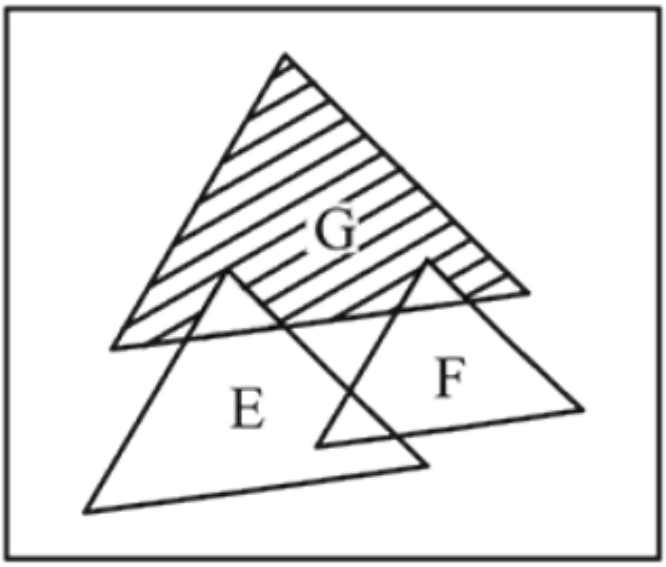
Q.7 Answer:

$\mathrm{P}\left(\mathrm{E}^{\mathrm{c}}\right)-\mathrm{P}(\mathrm{F})$

Solution:

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We have



$\because E \cap F \cap G = \phi$

$P\left(E^c \cap F^c / G\right) = \frac{P(E^c \cap F^c \cap G)}{P(G)}$

$= \frac{P(G) - P(E \cap G) - P(G \cap F)}{P(G)}$

[From venn diagram $E^c \cap F^c \cap G = G - E \cap G - F \cap G$**]**

$= \frac{P(G) - P(E)P(G) - P(G)P(F)}{P(G)}$

$= 1 - P(E) - P(F) = P\left(E^c\right) - P(F)$

[$\because E, F, G$ **are pairwise independent]**

Q.8 Answer:

$P\left(A'\right) + P\left(B'\right) = 2 + 2q - p$

Solution:

It is given that $P(A \cap B) = p$ **and** $P\left(A' \cap B\right) + P\left(A \cap B'\right) = q$

Therefore, since $P\left(A' \cap B\right) = P(B) - P(A \cap B)$

we get $q = P(B) - P(A \cap B) + P(A) - P(A \cap B)$

$\Rightarrow P(A) + P(B) = q + 2p$

$\Rightarrow P\left(A'\right) + P\left(B'\right) = 1 - P(A) + 1 - P(B)$

$= 2 - q - 2p$

showing that (b) is correct. The answer (c) is also correct because

$P(A \cap B \mid A \cup B) = \frac{P[(A \cap B) \cap (A \cup B)]}{P(A \cup B)} = \frac{P(A \cap B)}{P(A \cup B)}$

$= \frac{P(A \cap B)}{P(A) + P(B) - P(A \cap B)} = \frac{p}{q + 2p - p} = \frac{p}{p + q}$

Finally, (d) is correct because $P\left(A' \cap B'\right) = 1 - P(A \cup B)$

$= 1 - [P(A) + P(B) - P(A \cap B)]$

$= 1 - (q + 2p - p) = 1 - p - q$

Q.9 Answer:

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$1/3$

Solution:

Consider the following events:

A **: Father has at least one boy** B **: Father has 2 boys and one girl Then,** $A =$ **one boy and 2 girls, 2 boys and one girl,**

3 boys and no girl $A \cap B =$ **2 boys and one girl Now, the required probability is** $P(A/B) = \frac{P(A \cap B)}{P(A)} = \frac{1}{3}.$

Q.10 Answer:

$1/3$

Solution:

$\mathrm{P(A)} = 1/4, \mathrm{P(A/B)} = 1/2$ **and** $\mathrm{P(B/A)} = 2/3$

$P(B) =?$

$\mathbf{P}\left(\frac{\mathrm{B}}{\mathrm{A}}\right) = \frac{\mathrm{P(B \cap A)}}{\mathrm{P(A)}} \quad \frac{2}{3} = \frac{\mathrm{P(B \cap A)}}{1/4}$

$\Rightarrow P(B \cap A) = \frac{1}{6}$

$\mathbf{P}\left(\frac{\mathrm{A}}{\mathrm{B}}\right) = \frac{\mathrm{P(A \cap B)}}{\mathrm{P(B)}} = \frac{1/6}{\mathrm{P(B)}} \quad \frac{1}{2} = \frac{1}{6\mathrm{P(B)}} \quad \Rightarrow \mathrm{P(B)} = \frac{1}{3}$

Q.11 Answer:

$1 - \frac{1}{n!}$

Solution:

Probability of all the letters kept in the right envelope is

$\frac{1}{n!}$ $(\because$ **Total letters** $= n$ **) i.e.,** $\mathrm{P} = \frac{1}{n!}$

We know, if q **is the term used for the probability of the letters which are not kept in the right envelope.**

Then $p + q = 1 \Rightarrow q = 1 - p = 1 - \frac{1}{n!}.$

Q.12 Answer:

$\frac{1}{10}$

Solution:

The probability that the door is opened in the fifth trial $= \frac{9}{10} \times \frac{8}{9} \times \frac{7}{8} \times \frac{6}{7} \times \frac{1}{6} = \frac{1}{10}$

Q.13 Answer:

$9 : 6 : 4$

Solution:

Probability of getting a composite number is $2/6 = 1/3$

Probability that A **will win the game is** $\left(\frac{2}{3}\right)\left(\frac{1}{3}\right) + \left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{1}{3}\right)$

$+ \left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{1}{3}\right) + \ldots$

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$= \frac{\frac{1}{3}}{1-\frac{8}{27}} = \left(\frac{1}{3} \times \frac{27}{19}\right) = \frac{9}{19}$

Probability that B will win the game is $\left(\frac{2}{3}\right)\left(\frac{1}{3}\right) + \left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{1}{3}\right)$
$+ \left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{1}{3}\right) + \ldots$
$= \frac{\frac{2}{9}}{1-\frac{8}{27}} = \left(\frac{2}{9} \times \frac{27}{19}\right) = \frac{6}{19}$

Probability that C will win the game is $\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{1}{3}\right) + \left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{1}{3}\right)$
$+ \left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{1}{3}\right) + \ldots$
$= \frac{\frac{4}{27}}{1-\frac{8}{27}} = \left(\frac{4}{27} \times \frac{27}{19}\right) = \frac{4}{19}$

So required ratio is $9 : 6 : 4$.

Q.14 Answer:

$13 : 5$

Solution:

Let $E =$ **the event that horse** A **wins** $E_1 =$ **the event that jockey** B **rides horse** A $E_2 =$ **the event that jockey** C **rides horse** A **According to question odds in favour of** $E_1 = 2 : 1 \therefore P(E_1) = \frac{2}{3}$ **and** $P\left(\frac{E}{E_1}\right) = \frac{1}{6}$ **(Since, when** B **rides** A**, all six horses are equally likely to win)** $P(E_2) = 1 - P(E_1) = 1 - \frac{2}{3} = \frac{1}{3}$

and $P\left(\frac{E}{E_2}\right) = 3P\left(\frac{E}{E_1}\right) = \frac{1}{2}$

Let $A_1 = E_1 \cap E$ **and** $A_2 = E_2 \cap E$

Now, required probability $P(E) = P(A_1) + P(A_2)$
$= P(E_1 \cap E) + P(E_1 \cap E)$
$= P(E_1) P\left(\frac{E}{E_1}\right) + P(E_2) P\left(\frac{E}{E_2}\right)$
$= \frac{2}{3} \cdot \frac{1}{6} + \frac{1}{3} \cdot \frac{1}{2} = \frac{5}{18}$

So, that odds against winning of A **are** $13 : 5$.

Q.15 Answer:

$\frac{15}{2^{13}}$

Solution:

Let n **be the number of tosses and** X **the number of times heads occurs. Then** $\mathrm{X} \sim \mathrm{B}(\mathrm{n}, \mathrm{p})$**, with** $\mathrm{p} = 1/2$**. Therefore, since** $\mathrm{P}(\mathrm{X} = 7) = \mathrm{P}(\mathrm{X} = 9)$**, we have**
$\mathrm{C}_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^{\mathrm{n}-7} = {}^{\mathrm{n}}\mathrm{C}_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^{\mathrm{n}-9} \Rightarrow {}^{\mathrm{n}}\mathrm{C}_7 \left(\frac{1}{2}\right)^{\mathrm{n}} = {}^{\mathrm{n}}\mathrm{C}_9 \left(\frac{1}{2}\right)^{\mathrm{n}}$
That is, ${}^{\mathrm{n}}\mathrm{C}_7 = {}^{\mathrm{n}}\mathrm{C}_9 = {}^{\mathrm{n}}\mathrm{C}_{\mathrm{n}-9}$**, yielding** $7 = \mathrm{n} - 9$ **or** n
$= 16$**. Hence** $\mathrm{P}(\mathrm{X} = 2) = {}^{16}\mathrm{C}_2 \left(\frac{1}{2}\right)^{16} = \left(\frac{16 \times 15}{2}\right)\left(\frac{1}{2}\right)^{16} = \frac{15}{2^{13}}$

Q.16 Answer:

A and B are dependent

Solution:

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When a die is thrown, the sample space (S) is $S=\{1,2,3,4,5,6\}$
Let A : the number is even $=\{2,4,6\} \therefore P(A)=\frac{3}{6}=\frac{1}{2}$
B : the number is red $=\{1,2,3\} \therefore P(B)=\frac{3}{6}=\frac{1}{2}$
$A \cap B=\{2\}, P(A \cap B)=\frac{1}{6}$
or $P(A) \cdot P(B)=\frac{1}{2} \times \frac{1}{2}=\frac{1}{4} \neq \frac{1}{6}$

Q.17 Answer:

$\frac{7}{20}$

Solution:

Let E be the event that A speats the truth and let F be the event that B speaks the truth. The required probability is $P\left(EF^{\circ}\right)+P\left(E^{e}F\right)=P(E) \cdot P\left(F^{\circ}\right)+P\left(E^{\circ}\right) \cdot P(F)$ (where we have
assumed that A and B answer independently)
$=\frac{4}{5} \cdot \frac{1}{4}+\frac{1}{5} \cdot \frac{3}{4}=\frac{7}{20}$

Q.18 Answer:

$1/4$

Solution:

Since, India's second win occurs at the third test. Therefore, the sample space is $=[\mathrm{LWW}, \mathrm{WLW}]$
where, $\mathrm{L}=$ losing the test $\mathrm{W}=$ winning the test.
$\therefore \quad \mathrm{P}$ (India's win occur at the 3rd test $)=\mathrm{P}(\mathrm{LWW})+\mathrm{P}(\mathrm{WLW})$
$=\mathrm{P}(\mathrm{L})\mathrm{P}(\mathrm{W})\mathrm{P}(\mathrm{W})+\mathrm{P}(\mathrm{W})\mathrm{P}(\mathrm{L})\mathrm{P}(\mathrm{W})$
$(\because$ Prob. from match to match is independent).
$=\left(\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}\right)+\left(\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}\right)$ (given)
$=\frac{1}{8}+\frac{1}{8}=\frac{2}{8}=\frac{1}{4}$

Q.19 Answer:

E_1 and E_2 are exhaustive

Solution:

$$P\left(E_2 / E_1\right)=\frac{P\left(E_1 \cap E_2\right)}{P\left(E_1\right)}$$
$$\frac{1}{2}=\frac{P\left(E_1 \cap E_2\right)}{1/4} \quad \Rightarrow P\left(E_1 \cap E_2\right)=\frac{1}{8}$$
$$=P\left(E_2\right) \cdot P\left(E_1 / E_2\right)=P\left(E_2\right) \cdot \frac{1}{4}$$
$\Rightarrow$
$$\Rightarrow P\left(E_2\right)=\frac{1}{2} \text{ Since } P\left(E_1 \cap E_2\right)$$
$$=\frac{1}{8}=P\left(E_1\right) \cdot P\left(E_2\right)$$

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$\Rightarrow$ **events are independent Also** $\mathrm{P}\left(\mathrm{E}_1 \cup \mathrm{E}_2\right)=\frac{1}{2}+\frac{1}{4}-\frac{1}{8}=\frac{5}{8}$

$\Rightarrow \mathrm{E}_1 \& \mathrm{E}_2$ **are non exhaustive**

Q.20 Answer:

$17/28$

Solution:

Let A **be the event that** 1st **man will be alive till 75 years and** B **be the event that** 2nd **man will be alive till 70 years then** $\mathrm{P}(\mathrm{A})=5/16$ **and** $\mathrm{P}(\mathrm{B})=3/7$ **then** $\mathrm{P}\left(\mathrm{A}'\right)=11/16$**, and** $\mathrm{P}\left(\mathrm{B}'\right)=4/7$

Probability that none of them will be alive 35 years hence is $\mathrm{P}\left(\mathrm{A}'\right) \times \mathrm{P}\left(\mathrm{B}'\right)=11/16 \times 4/7=11/28$

Then required probability is $1-11/28=17/28$

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